

Task 1: To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;
    char buffer[100];

    fd = open("input.txt", O_RDONLY);

    if (fd == -1) {
        perror("input.txt");
        return 1;
    }

    dup2(fd, STDIN_FILENO);
    close(fd);

    while (fgets(buffer, sizeof(buffer), stdin) != NULL) {
        printf("%s", buffer);
    }

    return 0;
}
```

```
shreyas@Grigori: ~/redirection_tasks
GNU nano 7.2
Hello from input redirection.
This is the first line.
This is the second line.
```

```
shreyas@Grigori:~$ mkdir -p ~/redirection_tasks
shreyas@Grigori:~$ cd ~/redirection_tasks
shreyas@Grigori:~/redirection_tasks$ nano task1.c
shreyas@Grigori:~/redirection_tasks$ nano input.txt
shreyas@Grigori:~/redirection_tasks$ gcc task1.c -o task1
shreyas@Grigori:~/redirection_tasks$ ./task1
Hello from input redirection.
This is the first line.
This is the second line.
shreyas@Grigori:~/redirection_tasks$ rm input.txt
shreyas@Grigori:~/redirection_tasks$ ./task1
input.txt: No such file or directory
shreyas@Grigori:~/redirection_tasks$ echo "Testing input file again" > input.txt
shreyas@Grigori:~/redirection_tasks$ ./task1
Testing input file again
shreyas@Grigori:~/redirection_tasks$ ./task1 < input.txt
Testing input file again
```

Task 2: To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

```
GNU nano 7.2 task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt",
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

    if (fd == -1) {
        perror("output.txt");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("Hello from output redirection.\n");
    printf("This data is written into output.txt.\n");

    return 0;
}
```

```
shreyas@Grigori:~/redirection_tasks$ nano task2.c
shreyas@Grigori:~/redirection_tasks$ gcc task2.c -o task2
shreyas@Grigori:~/redirection_tasks$ ./task2
shreyas@Grigori:~/redirection_tasks$ cat output.txt
Hello from output redirection.
This data is written into output.txt.
shreyas@Grigori:~/redirection_tasks$ ls -l output.txt
-rw-r--r-- 1 shreyas shreyas 69 Sep 27 14:05 output.txt
shreyas@Grigori:~/redirection_tasks$ ./task2 > test_output.txt
shreyas@Grigori:~/redirection_tasks$ cat test_output.txt
Hello from output redirection.
This data is written into output.txt.
shreyas@Grigori:~/redirection_tasks$ chmod 000 output.txt
shreyas@Grigori:~/redirection_tasks$ ./task2
output.txt: Permission denied
shreyas@Grigori:~/redirection_tasks$ chmod 644 output.txt
shreyas@Grigori:~/redirection_tasks$ ls -l
total 48
-rw-r--r-- 1 shreyas shreyas 25 Sep 27 14:04 input.txt
-rw-r--r-- 1 shreyas shreyas 69 Sep 27 14:05 output.txt
-rwxr-xr-x 1 shreyas shreyas 16264 Sep 27 14:03 task1
-rw-r--r-- 1 shreyas shreyas 396 Sep 27 14:02 task1.c
-rwxr-xr-x 1 shreyas shreyas 16128 Sep 27 14:05 task2
-rw-r--r-- 1 shreyas shreyas 439 Sep 27 14:04 task2.c
-rw-r--r-- 1 shreyas shreyas 0 Sep 27 14:05 test_output.txt
shreyas@Grigori:~/redirection_tasks$ cat output.txt
Hello from output redirection.
This data is written into output.txt.
shreyas@Grigori:~/redirection_tasks$ cat test_output.txt
Hello from output redirection.
This data is written into output.txt.
shreyas@Grigori:~/redirection_tasks$ rm -f task1 task2 input.txt output.txt test_output.txt
shreyas@Grigori:~/redirection_tasks$ ls
task1.c task2.c
```